

The Role of Central Banks

For readers of a technical book like this, the relevant aspect of central bank activity is not so much why central banks act, but how and when they do it. Central banks are such large actors in the market that their operations affect various aspects of market microstructure. With the advent of unconventional monetary policy tools, central bank operations have extended along the yield curve and across asset classes. This chapter will spend only a minimum amount of space on the monetary policy aspect of central banks (what Bindseil [10] would call the ‘white collar’ part of central banking) and focus more on monetary policy tools and frameworks (the ‘blue collar’ aspect of their work). Readers looking for deeper insights into the policy aspects have access to a variety of good textbooks, including from practitioners like Bernanke [9], Issing [51] and Shirakawa [75].

5.1 INTRODUCTION

Central banks are banks that act as lenders and deposit takers to other banks. Most central banks also offer similar services to their own government (usually with tight restrictions or outright prohibitions on lending) and other customers, for instance foreign central banks or governments. While the history of central banks goes back several centuries, some countries came to central banking only late, notably the United States [3].

The designation central bank implies that these institutions play a fundamental role in their respective economy. Aside from this implication, however, there is not only no uniform definition of what exact function makes a bank a central bank, but also no consensus in economics about the necessity of central banks. The general public is also sometimes confused as to which institution are central banks. While the Bank of England and the Bank of Japan are central banks, Bank of Ireland and Deutsche Bank are private institutions¹. The National Bank of Slovakia is a central bank while the National Bank of Greece is not.

Ownership is also not a reliable guide. Most central banks are owned by sovereigns but not all. The US Federal Reserve Banks are privately owned, as is the Banca d’Italia,

¹The central banks of Ireland and Germany are the Central Bank of Ireland and the Deutsche Bundesbank, respectively. Bank of Ireland is government-owned at the time of writing.

and the Central Bank of Greece is listed on the Athens Stock Exchange. The nationalisation of the Bank of England in 1946 did not fundamentally change its position in the UK financial system.

A usual starting point for the definition of a central bank is that of a bank of issue, i.e., a bank that issues money. Bindseil [11] adopts this definition, with extra provisions so that a central bank is identified by three characteristics: it issues financial money of ultimate quality, it is based on a public charter, and it has been set up with the aim of public policy objectives. These three elements need further clarification, however. The money issued by the central bank need not be physical currency but can also be book entry (giro) money.

As Figure 5.1 shows, some currency systems exist today where bank notes are issued by private, non-central banks. Meanwhile, US Dollar notes, shown in Figure 5.2, are issued by the US Treasury, not the central bank.



FIGURE 5.1 A Hong Kong Dollar note issued by the Hongkong and Shanghai Banking Corporation (HSBC). The central bank of Hong Kong is the Hong Kong Monetary Authority (HKMA) which does not issue currency itself but governs the note issuance of private banks. Source: HSBC/HKMA.



FIGURE 5.2 A US Dollar note bears the seal of the Federal Reserve System but is issued by the US Treasury and signed by the Treasury Secretary. Source: US Treasury.

The Bindseil definition does include these two institutions when one considers the clearing accounts offered by both to the banks in their respective jurisdictions². Physical paper money is merely the form of money used by the general public but the bulk of transactions is conducted in electronic form.

Other authors, such as Goodhart [42] link the central bank definition less to the mechanical issuance of currency but to the aims of such issuance, which is generally linked to the provision of liquidity in times of stress, known as the lender of last resort (LOLR) function. The difference with the other definition is that this function is historically newer and so creates a somewhat arbitrary starting date for the emergence of central banks. As authors such as Bindseil [11] or Roberds and Velde [70] highlight, some central bank functions can be shown to have existed as early as the 15th century. Considering these institutions as central banks means that the emergence of the modern LOLR central banks (usually associated with the creation of the Swedish Riksbank, then known as Riksbens Ständers Bank) was an evolutionary step rather than a revolution. It is more plausible, particularly in the light of the tightly integrated trading relationships in Europe even in the late middle ages, that central banking developed gradually, one innovation at a time, across multiple countries over a longer period. In this approach, the Riksbank retains its claim to be the *oldest* central bank, but it is not the *first*.

One could therefore argue that the LOLR function is not always part of the charter of a central bank. The Banque de France and the Federal Reserve are examples of central banks that were chartered as LOLR banks (although for very different purposes), while the LOLR role of the Bank of England was only legislated after it had evolved *de facto*.

Whether one sees the LOLR function as the defining feature of a central bank, or takes the wider view of Bindseil, a central bank is defined via its liability side while the asset side is less specified. A central bank is therefore not necessarily a bank that funds the state, as is sometimes claimed. Indeed, the credibility of a central bank is often linked to its independence from the state, which underpins the separation between money and state debt.

Not all economists agree with this focus on the liability side. Funding of government expenditure, stability anchor for the financial system, and other roles compete for the ultimate reason for the existence of central banks in the literature.

This disagreement reflects the varied history of the institutions that make up the current line-up of central banks. Some of them were indeed created to ensure stable government funding, i.e., with an asset-side focus, like the Banque de France. Others were set up for the LOLR function in response to particular financial crisis, such as the Federal Reserve System in the United States. The Bank of England was set up with legal privileges, but its central function in the London financial market evolved in ways

²An additional quibble one might have with this definition is the precise nature of the phrase ‘public charter’. At the times when some of the early central banks were set up, there was no universal suffrage which means that a charter would be issued by a monarch or, in the city states, a body representing a small group of eminent citizens. The latter setting is not very different from a joint stock company. Many private banks were also set up with royal charters, for instance the first Pfandbrief issuer, Schlesische Landschaften, or Deutsche Bank.

that were not always predetermined by these legal arrangements. That being said, it is difficult to argue against the need for a publicly chartered LOLR bank given that any other currency arrangements (so-called free banking) has tended to end up in trouble.

As mentioned earlier, banking in Europe started from the business of taking deposits in the form of specie (precious metal) against the issuance of negotiable paper certificates, or the provision of transferrable giro credit. Because these paper certificates or giro credit could be exchanged back into specie on demand, they took on the role of money itself. However, the use of paper money in daily transactions meant that paper currency could remain in circulation instead of being frequently exchanged back into specie. This had the important side effect of leaving specie in circulation rather than tying it up in bank reserves. What made central bank-issued money different from any other monies in the economy is that the central bank money is of higher quality, either because it has legislative support, is more widely accepted, is convertible into specie of more reliable quality, or because the credit quality of the central bank is higher. In any case, holders of non-central bank money would prefer to convert it into central bank money.

The development of bank money requires two separate preconditions. First, the specie into which the bank money could be exchanged has to be widely accepted. This is a condition on the social contract valid at the time. The use of bank money does not in itself represent a change in currency. When bank money is used under a gold standard, gold remains the basic means of exchange. In the current fiat money regimes, private issue bank notes, where they exist, and private bank giro money merely represent, but do not replace, the basic means of exchange which is the liabilities of the central bank.

Second, because the obligation to exchange paper into specie rests with the issuing bank, users of the paper currency must have trust in the credit of that particular bank, i.e., its ongoing ability to convert its paper into specie on demand. This is all the more important in a system of fractional reserve banking where banks by construction are unable to redeem all outstanding notes for specie at the same time.

While this setup has evolved in the context of historical currency arrangements, it remains valid in today's world of fiat currencies and electronic transactions. Individuals are willing to be paid in the liabilities of individual private banks by accepting payments into their bank accounts. These bank accounts are not substantially different from paper bank notes issued by banks in the past. Individuals can also use these same bank liabilities to make payments when they transfer money electronically.

The second condition is more complex when one considers a banking system comprised of many banks rather than a few big names. Users of paper currency issued by multiple banks need to trust every single issuing bank. In practice, this has meant that only certain types of private paper money were accepted widely, leading to the natural emergence of oligopolies.

The need for the existence of a central bank and a privileged legal tender is sometimes disputed, as is the creation of money by private banks [42]. On one extreme, proponents of free banking suggest that private banks should be allowed to issue bank notes freely and that the role of central banks should be limited [44]. In this model,

the users of currency would be deciding which banks to trust, which would then put pressure on bank managers to behave in a demonstrably prudent way³.

On the opposite side of free banking is the idea of positive money, called *Vollgeld* in Germany, where this theory has some following. In this type of banking model, private banks do not have the right to create money. Instead, they need to borrow any money they wish to advance as loans from the central bank. While central bank money retains the features of a fiat currency (the central bank would have the right to issue as much of its money as it sees fit), the amount of money in circulation would be decided by the state, not private actors. Banks would be mostly providers of payment and other services and any risky lending activity would in essence mirror the operation of asset managers. If one allows for the assumption that it does not matter which private bank advances a particular loan, positive money is merely the extreme where the amount of total credit in the economy is exactly controlled by the government. As will be discussed later on, central banks can influence the amount of private money creation in various ways even not as precisely as positive money advocates would view as desirable. The Bundesbank operated under a regime that targeted a given rate of money supply growth⁴ and the US Federal Reserve had tight money supply targets during its Friedmanite episode between 1979 and 1982. Some economies experienced government targets for credit growth, for instance the window guidance employed in Japan in the 1980s and China in the early 2000s [38].

Giving authority to decide on money creation to the largest borrower in the economy (the government) is seen as sensible on the grounds that governments are appointed by the collective wisdom of the population through democratic elections⁵. Control over the money supply, which can be implemented through positive money, could allow the government to reduce the cost of its debt by conducting monetary policy in such a way as to reduce the function of money as a store of value. While this use of monetary policy for the augmentation of government finances is also possible in fractional reserve banking systems [28], positive money eliminates the alternative of private monies.

It should not be surprising that the relative popularity of these schools of thought is somewhat cyclical. When the economy is doing well, private actors feel less inclined to submit to government oversight. Free banking then tends to be advocated as a natural expression of private enterprise. Conversely, in a recession many people wish

³The idea that publishing a bank's situation encourages good management can be seen in pillar III of the Basel framework.

⁴Some authors argue that while money supply may have been the stated target, actual policy measures were more in line with inflation targeting [28], or at least on the road to it [9].

⁵It should be noted in this context that the fundamental ideas of positive money actually predate modern democracies, cf. for instance the creation of the *Banque de France* under Napoleon [42].

for more state intervention and positive money concepts become more popular⁶. The co-existence of private monies with a controlling role of central bank monies in most modern economies is a compromise between these two extremes.

5.2 MONETARY FINANCING

The term monetary financing describes the creation of money with the aim of funding fiscal expenditure and is older than central banking. A government typically raises money from the populace through taxes, levies, and tariffs. On the back of these revenue streams, the government can also borrow money, and, as pointed out elsewhere, historically these revenue sources were not as clearly separated as they are today [57].

In a metal standard currency, monetary financing does not require that the government issues money that is legislated as a means of exchange (forced money), or forces the central bank to issue money on its behalf. The same economic effect has historically been achieved by diluting the specie content of a standard coin (debasement), or by seizing the specie reserves of a money-issuing bank. In all of these cases, the precious metal amount that a given monetary unit can be converted into will be less than expected, and may even be zero. The problem is that a government, and in particular a non-democratic one, can override the public interest in stable money.

This loss of conversion value is why monetary financing is problematic for money because it impedes the store-of-value function. In medieval times, monetary financing was in some cases avoided by sheltering the money-issuing central bank in a religious institution [11]. Today's approach is to make central banks independent and rule out monetary financing by law. Central banks are either completely forbidden from lending to their sovereigns (Europe) or can only do so to a very limited degree (US and Japan). Where monetary financing is not expressly forbidden in law, it tends to be ruled out as a matter of policy because it clashes with the aim of maintaining stable money.

Practically all central banks are authorised to buy and sell bonds issued by their sovereigns and many do so as part of their operational framework. As will be discussed below, non-standard monetary policy tools require large-scale purchases of sovereign debt. Because a holder of a debt instrument acts as a lender to the issuer of that instrument, the question arises how such holding can be reconciled with a prohibition on monetary financing.

⁶It tends to be forgotten in these arguments that government-controlled banks can collapse as well as private ones. To a larger extent than the more general criticisms of central planning (e.g. [44]), any default of a public bank should create severe doubts about the ability of governments to police the financial system. Planning a whole economy centrally is a computationally complex task and failure to do so successfully in the former Eastern bloc can be explained away by this complexity. Running a German or Austrian Landesbank, in contrast, is not evidently more complex than running a private bank. The observed cases of failures to do so better than private management cannot as easily be separated from general control abilities in the public sector.